Total No. of Questions : 8]

SEAT No. :

PA-913

[Total No. of Pages : 3

[5927]-343

B.E. (Computer)

MACHINE LEARNING

(2019 Pattern) (Semester - VII) (410242)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Make suitable assumption whenever necessary.***

Q1) a) Explain in brief techniques to reduce under fitting and over fitting. **[6]**

b) Find the Equation of linear Regression line using following data : **[6]**

X	Y
1	3
2	4
3	5
4	7

c) Write short note on : **[6]**

i) MAE

ii) RMSE

iii) R^2

OR

Q2) a) Explain in brief lasso and Ridge Regression. **[6]**

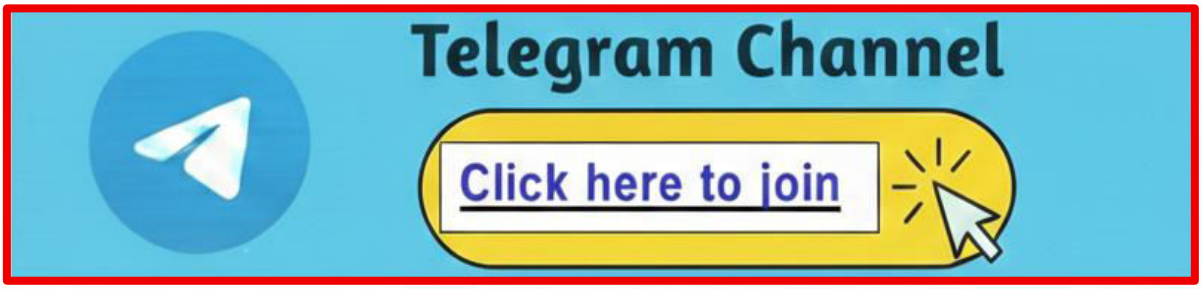
b) What is Bias and variance trade off for machine learning model? **[6]**

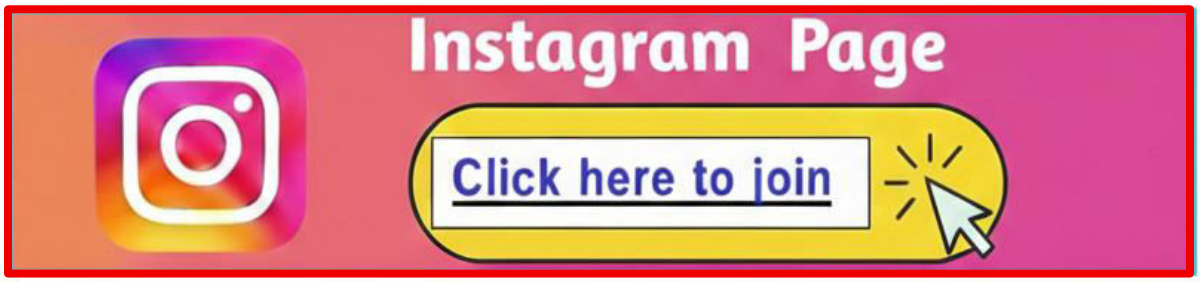
c) Write short note on Evaluation metrics. **[6]**

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Q3) a) Explain in brief methods used for Evaluating classification models. **[5]**

b) Consider the following data to predict the student pass or fail using the K-Nearest Neighbor Algorithm (KNN) for the values physics = 6 marks, Chemistry = 8 marks with number of Neighbors K = 3. **[6]**

Physics (marks)	Chemistry (marks)	Results
4	3	Fail
6	7	Pass
7	8	Pass
5	5	Fail
8	8	Pass

c) Write short note on Ensemble learing methods : **[6]**

i) Simple

ii) Advanced

OR

Q4) a) Explain Random forest Algorithm with example. **[5]**

b) Write short note on importance of confusion matrix. **[6]**

c) Define following terms with reference to SVM. **[6]**

i) Separating hyperplane

ii) Margin

Q5) a) Explain Density Based clustering with refence to DBSCAN, OPTICS and DENCLUE. **[6]**

b) What is K mean clustering? Explain with example. **[6]**

c) Write short note on following Hierarchical clustering method : **[6]**

i) Agglomerative

ii) Dendogram

OR

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Q6) a) What is LOF? Explain it with it's advantages and disadvantages. **[6]**

b) Explain Graph Based clustering. **[6]**

c) Define following terms : **[6]**

i) Elbow method

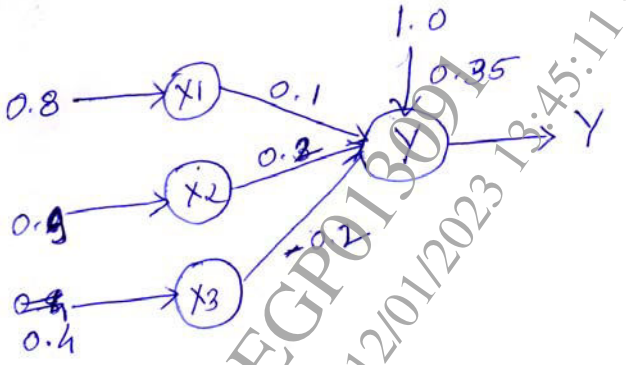
ii) Extrinsic and Intrinsic method

Q7) a) Explain ANN with it's Architecture. **[5]**

b) Obtain the output of Neuron Y for the Network shown in following fig. Using activation function as : **[6]**

i) Binary sigmoidal

ii) Bipolar sigmoidal



c) Write short note on Back propagation network. **[6]**

OR

Q8) a) Explain in brief types of ANN based on layers. **[5]**

b) What is Recurrent Neural Network? Explain with suitable example. **[6]**

c) Write short note on with refernce with CNN. **[6]**

i) Convolution layer

ii) Hidden layer

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Total No. of Questions : 8]

SEAT No. :

P546

[Total No. of Pages : 2

[6004]-481
B.E. (Computer Engineering)
MACHINE LEARNING
(2019 Pattern) (Semester - VII) (410242)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.***
2) ***Figures to the right indicate full marks.***
3) ***Neat diagrams must be drawn wherever necessary.***
4) ***Assume suitable data, if necessary.***

Q1) a) Explain the following terms with suitable examples. **[6]**

i) Bias

ii) Variance

iii) Under fitting and Over fitting

b) Differentiate between Lasso Regression and Ridge Regression. **[6]**

c) Explain gradient descent algorithm with example. **[6]**

OR

Q2) a) What do you mean by regression? Explain with suitable example. **[6]**

b) Write a short note on : **[6]**

i) MAE

ii) RMSE

iii) R^2

c) What is gradient descent? Compare batch gradient and stochastic gradient descent. **[6]**

Q3) a) Explain with example the variant of SVM, the support vector regression. **[5]**

b) What do you mean by ensemble learning? Differentiate between bagging & boosting. **[6]**

c) What are different variants of multi-class classification? Explain them with suitable example. **[6]**

OR

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Q4) a) Calculate macro average precision, macro average recall and macro average F-score for the following given confusion matrix of multi-class classification. **[6]**

Predictions →

Actual values ↓	A	B	C	D
A	100	80	10	10
B	0	9	0	1
C	0	1	8	1
D	0	1	0	9

b) Write a short note on : **[6]**

i) Random forest.

ii) Adaboost.

c) Discuss K-nearest neighbour algorithm with suitable example. **[5]**

Q5) a) With reference to Clustering explain the issue of "Optimization of Clusters". **[6]**

b) Compare Hierarchical clustering and K-means clustering. **[6]**

c) Explain how a cluster is formed in the density based clustering algorithm. **[6]**

OR

Q6) a) How would you choose the number of clusters when designing a K-Medoid clustering algorithm? **[6]**

b) Write a short note on out lier analysis with respect to clustering. **[6]**

c) Differentiate between K-means and Spectral clustering. **[6]**

Q7) a) What are building blocks of neural network, elaborate? **[5]**

b) Describe characteristics of back propagation algorithm. **[6]**

c) Write a short note on Recurrent neural n/w & convolutional neural n/w. **[6]**

OR

Q8) a) Explain artificial neural n/w based on perception concept with diagram. **[6]**

b) Describe multi-layer neural n/w. Explain why back propagation algorithm is required. **[6]**

c) Discuss any two activation functions with example. **[5]**



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Total No. of Questions : 8]

SEAT No. :

P-6552

[Total No. of Pages : 2

[6181]-102

B.E. (Computer Engineering)

MACHINE LEARNING

(2019 Pattern) (Semester - VII) (410242)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right side indicate full marks.***
4) ***Assume suitable data if necessary.***

Q1) a) Differentiate between overfitting and underfitting. **[6]**

b) The table below shows the number of grams of carbohydrates, X and the number of Calories, Y of six different foods. Find linear regression equation for this dataset. **[8]**

Carbohydrates (X)	8	9.5	10	6	7	4
Calories (Y)	12	138	147	88	108	62

Also find the value of Y for X = 12

c) Explain Bias Variance Trade off. **[4]**

OR

Q2) a) What is Linear Regression? Explain the concept of Ridge regression. **[9]**

b) Explain the following Evaluation Metrics : **[9]**

i) MAE

ii) RMSE

iii) R2

Q3) a) Differentiate between bagging and boosting. **[4]**

b) What is ensemble learning? Explain the concept of Random Forest ensemble learning. **[9]**

c) What is the relation between precision and recall? Explain with an example. **[4]**

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OR

Q4) a) What is K-fold cross-validation? In K-fold cross-validation, comment on the following situations **[9]**

i) When the value of K is too large

ii) When the value of K is too small.

How do you decide the value of k in k-fold cross-validation?

b) Explain i) Accuracy, ii) Precision, iii) Recall, and iv) F-Score **[8]**

Q5) a) Explain K-Means clustering in detail with a suitable example. **[8]**

b) What is outlier analysis? How is Local Outlier Factor detected? **[5]**

c) Explain Spectral Cluster in galgorithm. **[5]**

OR

Q6) a) Explain Hierarchical and Density-based Clustering approaches. **[9]**

b) Write short note on : **[9]**

i) Optimization of clusters

ii) K-Medoids

iii) Evaluation metrics

Q7) a) Write a note on Single Layer Neural Network. **[4]**

b) Explain Radial Basis Function networks in detail. **[8]**

c) Explain Recurrent Neural Networks and its applications in brief. **[5]**

OR

Q8) a) Explain the concept of Back Propagation in ANN with example. **[8]**

b) What is Functional Link Artificial Neural Network (FLANN)? Explain its merits over other ANNs. **[5]**

c) What is Activation Function? Explain with a suitable example. **[4]**

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Total No. of Questions : 8]

SEAT No. :

PB-2244

[Total No. of Pages : 2

[6263]-82
B.E. (Computer Engineering)
MACHINE LEARNING
(2019 Pattern) (Semester - VII) (410242)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Figures to the right side indicate full marks.***
3) ***Draw neat diagram wherever necessary.***
4) ***Assume suitable data, if necessary.***

Q1) a) Define different regression models. **[6]**

b) What are different techniques to reduce under fitting? **[6]**

c) With following data of shows company expenditure. **[6]**

x(month)	1	2	3	4	5
y(expenditure)	12	19	29	37	45

using regression model predict expenditure of 6th month.

OR

Q2) a) What is R2 measure of evaluation? **[6]**

b) What do you mean by least square method? Explain least square method in the context of linear regression. **[6]**

c) Write a short note on stochastic qradient descent algorithms. **[6]**

Q3) a) Why ensemble learning is used for ML? **[5]**

b) What are advantages and disadvantages of K-NN? **[6]**

c) What are different distance metrics used in k-NN? **[6]**

OR

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Q4) a) What is multiclass classification? Explain the variants of multiclass classification. **[5]**

b) Explain kernel methods which are suitable for SVM. **[6]**

c) What are different techniques used for outlier handling? **[6]**

Q5) a) Why K-medoid is used? Explain k-medoid algorithm. **[5]**

b) Why density based clustering is used? Explain any one. **[6]**

c) What is outlier analysis? **[6]**

OR

Q6) a) What is isolation factor model? **[5]**

b) Explain k means algorithm. **[6]**

c) Explain Hierarchical clustering with example **[6]**

Q7) a) What is Multilayer perceptron? Describe with diagram. **[6]**

b) What are different activation function used is NN? **[6]**

c) Explain Convolution Neural Network. (CNN) with suitable example. **[6]**

OR

Q8) a) Explain building blocks of RBF networks. **[6]**

b) What is Personalized recommendation? What is content based recommendation? **[6]**

c) Explain Recurrent Neural Networks with as example. **[6]**

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Total No. of Questions : 8]

SEAT No. :

PC-2369

[Total No. of Pages : 2

[6354]-486

B.E. (Computer Engineering)

MACHINE LEARNING

(2019 Pattern) (Semester - VII) (410242)

Time : 2 ½Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 OR Q.2, Q.3 OR Q.4, Q.5 OR Q.6, Q.7 OR Q.8.***
2) ***Figures to the right side indicate full marks.***
3) ***Draw neat diagram wherever necessary.***
4) ***Assume suitable data, if necessary.***

Q1) a) Explain Lasso Regression. Explain how Lasso Regression is used for feature selection. **[6]**

b) Define different regression models. **[6]**

c) Describe the bias-variance trade-off and its relationship to under fitting and overfitting. **[6]**

OR

Q2) a) Explain the advantages of RMSE over MSE as an evaluation metric. **[6]**

b) What do you mean by least square method? Explain least square method in the context of linear regression. **[6]**

c) Write a short note on Stochastic gradient descendent algorithms. **[6]**

Q3) a) Explain kernel methods which are suitable for SVM. **[6]**

b) What are advantages and disadvantages of K-NN? **[6]**

c) What are different distance metrics are used in K-NN? **[5]**

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OR

Q4) a) What is Multi Class Classification? Explain the variants of Multi Class Classification. **[5]**

b) What are different techniques used for outlier handing? **[6]**

c) With suitable diagram, Explain Random forest Algorithm with example. **[6]**

Q5) a) Why K-medoid is used? Explain K-medoid algorithm. **[5]**

b) Why density based clustering is used? Explain any one. **[6]**

c) Cluster the following eight points (with (x, y) representing locations) into three clusters: **[6]**

P1(1, 3), P2(2, 2), P3(5, 8), P4(8, 5), P5(3, 9), P6(10, 7), P7(3, 3), P8(9, 4), P9(3, 7)

Use K-Means Algorithm to find the three cluster

OR

Q6) a) What is isolation factor model? **[5]**

b) Explain Hierarchical Clustering with an example. **[6]**

c) Micro-Average Precision and Recall, Micro-Average F-score, **[6]**

Q7) a) Explain Recurrent Neural Networks with an example. **[6]**

b) What are different activation function used in NN? **[6]**

c) What is multilayer perceptron? Describe with diagram. **[6]**

OR

Q8) a) Explain building blocks of RBF networks. **[6]**

b) What is personalized recommendation? What is content based recommendation? **[6]**

c) Explain the Convolution Neural Network (CNN) with suitable example. **[6]**



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Total No. of Questions : 8]

SEAT No. :

PD4577

[Total No. of Pages : 2

[6404]-82
B.E. (Computer Engineering)
MACHINE LEARNING
(2019 Pattern) (Semester - VII) (410242)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Figures to the right indicate full marks.***
3) ***Neat diagrams must be drawn wherever necesssary.***
4) ***Make suitable assumption whenever necessary.***

Q1) a) Explain Linear Regression in brief. **[6]**

b) Differentiate overfitting and under fitting with example. **[6]**

c) Write short note on Training error and Generalization error in machine learning. **[6]**

OR

Q2) a) Explain in brief importance of Evalution matrix. **[6]**

b) Write short note on Lasso and Ridge Regression. **[6]**

c) Memory (Capacity) and cost of RAM as shown in below table. **[6]**

X (Memory Capacity) in GB	2	4	8	16
Y (Cost in $)	12	16	28	62

i) Find Regression line $Y = aX + b$ using least square method.

ii) Estimate the cost of 32 GB RAM using line as equation.

Q3) a) Differentiate Binary classification with multiclass classification with example. **[5]**

b) Explain with example ensemble learning in ML. **[6]**

c) Write short note on metrics for Evaluating classifier performance. **[6]**

OR

Q4) a) Explain KNN with example. **[5]**

b) Write short note on handling imbalance data in multiclass classification. **[6]**

c) Explain Bagging and Boosting used in ensemble learning. **[6]**

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Q5) a) Explain Hierarchical clustering with example. **[6]**

b) What is outlier Analysis? Explain it with importance, advantages & disadvantages. **[6]**

c) Write short note on Elbow method used in K-mean clustering. **[6]**

OR

Q6) a) Compare Intrinsic motivation with extrinsic motivation. **[6]**

b) Explain K mean clustering with essential steps used in it. **[6]**

c) Write short note on. **[6]**

i) Graph Based clustering

ii) Density Based clustering

Q7) a) Write short note on importance of Activation function used in Neural Network. **[5]**

b) Explain Recurrent Neural Network with Example. Compare it simple CNN. **[6]**

c) Draw & Explain CNN architecture. **[6]**

OR

Q8) a) Compare Back propagation Network with feed forward Network. **[5]**

b) What are different types of padding used in CNN? **[6]**

c) Write short note on layers used in CNN. **[6]**

i) Convolutional layer

ii) Pooling layer

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Total No. of Questions : 8]

SEAT No. :

PE-2193

[Total No. of Pages : 3

[6584]-92
B.E. (Computer Engineering)
MACHINE LEARNING
(2019 Pattern) (Semester - VII) (410242)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) ***Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.***
2) ***Figures to the right side indicate full marks.***
3) ***Draw neat diagram wherever necessary.***
4) ***Assume suitable data, if necessary.***

Q1) a) What is R2 measure of evaluation? **[6]**

b) Explain the concept of a cost function and its role in Gradient Descent. **[6]**

c) What is the primary purpose of the L2 regularization term in the Ridge Regression cost function? **[6]**

OR

Q2) a) Describe the bias-variance trade-off and its relationship to underfitting and overfitting. **[6]**

b)

Month	Amount spent	Sales
Mar-23	13447.45	5750.00
Feb-23	10692.38	44305.00
Jan-23	3067.05	34980.00
Dec-22	200.00	10500.00
Nov-22	12343.18	39154.00
Oct-22	12300.18	23065.00
Sep-22	11960.35	20745.00
Aug-22	9924.53	42000.00
Jul-22	8873.12	5750.00
Jun-22	6383.40	41500.00
May-22	1397.60	13500.00

The table shows advertisement spent and sales figure for 11 months. Predict what would be sales if ad spend is 10800/- using linear regression. **[6]**

c) What are different techniques to reduce overfitting? **[6]**

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Q3) a) Explain the basic principles of a Support Vector Machine (SVM). What are support vectors and how does SVM find the optimal hyper plane for classification? **[5]**

b) What is Multi Class Classification? Explain the variants of Multi Class Classification. **[6]**

c) You have a binary classification model with the following confusion matrix: **[6]**

True Positives (TP): 120

True Negatives (TN): 45

False Positives (FP): 10

False Negatives (FN): 5

Calculate the accuracy, precision, recall, and F1-score of the model.

OR

Q4) a) With suitable diagram, Explain Random forest Algorithm with example. **[5]**

b) What are different distance metrics are used in K-NN? **[6]**

c) Why ensemble learning is used for ML? **[6]**

Q5) a) What is K-means clustering, and how does it work? Explain the basic principles of this unsupervised machine learning algorithm and its applications. **[5]**

b) Why K-medoid is used? Explain K-medoid algorithm. **[6]**

c) Define the terms Core point, Border point, Noise point, Eps (ε), MinPts (min - samples), Reachability, Density-connected used in the Density-Based Spatial Clustering (DBSCAN) algorithm. **[6]**

OR

Q6) a) What is isolation factor model? **[5]**

b) Write short note on following Hierarchical clustering method : **[6]**

i) Agglomerative

ii) Dendogram

c) Explain the intrinsic and extrinsic methods for measuring clustering quality. **[6]**

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Q7) a) Describe the architecture of Artificial Neural Network. **[6]**

b) Explain the Recurrent Neural Network (RNN) with suitable example. **[6]**

c) Explain building blocks of RBF networks. **[6]**

OR

Q8) a) What is multilayer perceptron? Describe with diagram. **[6]**

b) Explain the Convolution Neural Network (CNN) with suitable example. **[6]**

c) What is the role of activation functions in neural networks, and why are they necessary for the network's learning process? **[6]**

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